Theoretical Question (Idea)

Lecture 01

1. Identify and briefly explain the three major classes of applications for measurement systems.
2. List the three core functional elements of a measurement system and describe the primary purpose of each element.
3. Why do raw sensor outputs often require a signal conditioning element? Give two examples of signal conditioning processes.
4. Differentiate between active sensors and passive sensors, and provide a real-world example of each.
5. What is meant by the "calibration" of a sensor, and what does a calibration curve represent?
6. Define gross errors and systematic errors. Provide one example and one mitigation method for each type of error.
7. What causes random errors (noise) in a measurement system, and what is the standard method used to minimize their effect?
8. Distinguish clearly between accuracy and precision in measurement systems, detailing which type of error affects each property.

Lecture 02

1. Differentiate between indicating, recording, and integrating instruments, providing a specific example for each category.
2. Explain the two major technical disadvantages or risks associated with a simple multi-range ammeter switching arrangement.
3. Recall the significance of shunt resistance in an ammeter?
4. What is an Ayrton shunt (or universal shunt)? Explain how it successfully eliminates the structural limitations of a simple multi-range ammeter.
5. Explain what "meter loading" means in the context of an ammeter and how it alters the circuit being tested.
6. What is the structural and mathematical purpose of connecting multiplier resistors in series with a basic PMMC movement?
7. Describe the internal resistance property of an ideal voltmeter and explain why perfect voltmeters only exist in textbooks.

Lecture 03

1. What are the two primary technical disadvantages of a basic half-wave rectifier AC meter circuit? How to overcome this?
2. Explain how adding a second diode (D2) in a half-wave rectifier AC meter layout protects the movement and minimizes Peak Inverse Voltage (PIV) requirements.
3. Why do rectifier-based AC meters suffer from non-linearity at lower RMS voltage ranges, and how can this non-linearity be reduced?

Lecture 04

1. Under what conditions should a wattmeter's Potential Coil (PC) be connected to point ‘A’ (source side) versus point ‘B’ (load side) to minimize connection errors?
2. What are instrument transformers, and what parameters do they help measure in high-power applications?
3. State the two primary safety and engineering functions served by utilizing Current Transformers (CT) and Potential Transformers (PT).
4. Explain the construction and operating principle of a Current Transformer (CT). Analyze why the secondary winding of a Current Transformer (CT) should never be left open while the primary is energized.
5. Why do high-voltage Potential Transformers demand heavy insulation layouts between their primary and secondary windings compared to Current Transformers?

Lecture 05

1. Define "Transduction" and "Transducer." Explain the major advantages of using electrical transducers over mechanical methods.
2. Differentiate between a primary transducer and a secondary transducer.
3. Explain the underlying working principle of a Resistance Temperature Detector (RTD). What does a positive temperature coefficient (PTC) indicate?
4. List three critical material characteristics that an industrial-grade metal must possess to be highly effective as an RTD element.
5. Define a thermistor. Compare its temperature coefficient, sizing, sensitivity, and typical material compositions with those of standard metal RTDs.